
Vaccination And Small-Pox

Author(s): A. B. Steele

Source: *The British Medical Journal*, Vol. 1, No. 535 (Apr. 1, 1871), pp. 351-353

Published by: BMJ

Stable URL: <https://www.jstor.org/stable/25229282>

Accessed: 17-04-2025 17:02 UTC

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VACCINATION AND SMALL-POX.

VACCINATION AND REVACCINATION.

AN Irish dispensary medical officer of much experience has sent us the following practical remarks.

The question of vaccination and revaccination is one that is attracting considerable attention at the present time; and not without cause, as the present epidemic of small-pox, in England at least, is one of unusual severity. For this reason I venture to obtrude the results of my individual experience on the notice of your numerous readers. On referring to my vaccination registers, I find that I have vaccinated and revaccinated over five thousand individuals; those revaccinated probably numbering about one-tenth of the whole. I know that there are many others of my brother Poor-law medical officers who might go back on much greater experience and much larger numbers than these, and who could probably throw more light on this *vexata quæstio* than any other practitioners.

When the question is properly threshed out, as it is in process of being now, both in England and Ireland, it will be found that the Poor-law medical officers of Ireland and the public vaccinators in England will be the persons best capable of giving an opinion as to the great value of primary vaccination and how long its influence lasts. With regard to revaccination, the medical officers connected with the army and navy would, if they would give their experience, be most likely to enlighten us on this subject, as the great majority of men entering the services are revaccinated. As to primary vaccination, I think it would be utterly absurd to question its necessity; and with regard to revaccination, from my own experience and that of those most competent to form an opinion on the subject with whom I have spoken, or whose views have obtained publicity, I would certainly revaccinate, or recommend revaccination, to all persons who had not been revaccinated for a period of seven years, provided that their health was good at the time.

Vaccination will not, indeed, give complete immunity from small-pox; but, during the last epidemic of that disease in Ireland, seven years ago, I had more than 100 cases under my care, and I never yet saw a person die of it who had a well-marked vaccination cicatrix, but I have seen a person die who had well-marked small-pox cicatrices. For this reason, I have come to the conclusion that a successful vaccination is as good a preventive against small-pox as a previous attack of small-pox itself.

When is the best time to vaccinate? I have vaccinated at all ages. I generally recommend as soon as possible after two months old; because as children grow older they are more restless, and more likely to scratch their arms, which in both young children and adults is one of the principal causes of sore arms. I was once called to see a woman who had been confined the night before. While there, I discovered that another of her children was in bed in the same room, and had small-pox. He had been vaccinated, and the disease was (consequently, I would say) very mild. I immediately vaccinated the infant; the vaccination was successful. It took small-pox also, but, the vaccination having the start, the small-pox was very mild; one or two pustules alone appearing. The child is now a fine little girl, and her mother insisted on her being revaccinated lately. This was done with good result. Several cases similar to this have occurred in my practice, though not in such young children; and I am inclined to agree with those who say that, in a person successfully vaccinated on the day which he is exposed to small-pox, the vaccination will outstrip the variola, and so modify it that the most favourable results may be anticipated.

A great deal has been said about erysipelas, sore arms, and so forth, after revaccination. I have seen all these in children, both in primary vaccinations and in revaccinations; but they are the exceptions, and for obvious reasons are more likely to occur amongst adults, who are of course more exposed to external violence and constitutional disturbance than children, and who are more likely to bring the nature of the case before the notice of the public and the profession. I have often remarked that it is from surgeons and physicians who perhaps have vaccinated but very few persons in their lives who say they have seen such results attendant on revaccination; the fact being, that the half-dozen bad arms that arose after revaccination in unhealthy persons filtered through the crowd of the revaccinated until they accidentally came before their notice in hospital or otherwise, whereas nothing was heard of the thousand and one who were revaccinated without any bad results. As to the real advantage of revaccination, I will mention one case that came under my notice. I was once called in to attend the family of a relative of mine. They were all grown up, and at the time there were

several visitors at the house. My professional attendance was required for one of the sons, a lad aged 19. I immediately pronounced the case to be one of small-pox; he had been vaccinated, and it was modified. I recommended immediate revaccination for every other member of the household. They all agreed, with the exception of the mistress of the house, and I revaccinated them forthwith. Although they were almost all equally exposed to the influence of the disease, not one took it, with the exception of the lady who refused to be revaccinated. Having been vaccinated before, however, her case, though tedious and troublesome, was also modified, and she recovered without having more than three or four marks on her face. I then revaccinated myself and every member of my own family over seven years of age without the slightest bad result. Having revaccinated over five hundred individuals without, I can conscientiously add, any worse results to themselves than slight inconvenience, I believe that revaccination is desirable, and is also necessary when an epidemic is present or the individual is exposed to small-pox. The reports of the nurses of the London Small-pox Hospital appear to me to prove this conclusively. I do not think that it is clearly established that vaccination wears out; but I have myself taken, and I always give my patients, the benefit of the doubt, and revaccinate, of course taking into consideration the health of the individual at the time. In fact, it is not altogether optional with us in Ireland to revaccinate, for in Article 21 of the general rules for the Government of dispensary districts, under the head of "Duties of medical officers, section 8", it says: "He shall vaccinate all persons who may come to him for that purpose"; and a form is provided for those born before 1864. Still we do not vaccinate persons in a bad state of health; and I do not think that the Commissioners would insist on the literal acceptance of the section. As to the mode of vaccination, I believe two insertions to be sufficient; they have much less chance of being rubbed or scratched than a greater number. Vaccination has been performed on the leg, wrist, in the neighbourhood of the axilla, and other places; but the natural hollow in the arm at the insertion of the deltoid I think the best place. The two insertions I make one above the other, about an inch apart. When they are made horizontally, that on the outer surface of the arm is liable to be lain upon, and so injured. Vaccination should be performed on the arm which is not habitually lain upon, or the arm which is held outermost by the mother or nurse. The lancet should be kept scrupulously clean, and not used for any other purpose. The less blood drawn, the better. The scratches should be made horizontally, with the point of the lancet directed downwards, which gives them a sort of valvular effect. With regard to the storage of vaccine lymph, I have tried Husband's capillary tubes, glass plates, and ivory points, and I have no hesitation in saying that the ivory points are the best. Perhaps for a long voyage or in a tropical region the capillary tubes might be better, but of this I have no experience. The capillary tubes are troublesome to fill, and, unless carefully manipulated, the lymph will be boiled in the sealing. Although our Poor-law Commissioners are strongly in favour of them, I do not think that they are as good as ivory points, and, as a rule, we do not use them.

The foregoing remarks are merely the result of my individual experience, and do not contain anything new; but I have observed, in reading many reports which have appeared in the papers, both lay and medical, remarks which in my estimation exhibited a want of knowledge both of vaccination and of revaccination.

SMALL-POX AT CROYDON.

THE Croydon Board of Guardians, at their meeting on the 13th instant, resolved to appoint a medical officer to attend specially on the small-pox patients in the parish of Croydon and in the Small-pox Hospital of the town. The object of this arrangement is to avoid the risk of conveyance of infection by the other Poor-law medical officers of the union.

IS ONE REVACCINATION A PERMANENT PROTECTION AGAINST SMALL-POX?

MR. M. A. WOOD of Ledbury writes on this question as follows.

I have been expecting to see in the columns of this JOURNAL some notice taken of the statement of the Medical Officer of the Privy Council, that "revaccination once properly and successfully performed does not appear ever to require repetition". If this conclusion be not founded on sufficient evidence, it is due to the profession and also to the public (who will use it as an excuse for their not being revaccinated a second time) that it should not pass unchallenged. Is it reasonable to suppose that an immunity that can be lost once cannot be lost a second time? We know that in some eruptive fevers, and in small-pox itself, the disease does sometimes occur in the same individual more than

twice; and the experience of medical men must, I think, have shown that revaccination can be successfully performed more than once. The ground of Mr. Simon's argument seems to be that all the nurses and servants that have been attached to the Small-pox Hospital during the last thirty-four years, and who were properly revaccinated before their admission, but not since, have without exception escaped taking small-pox. This striking fact appears at first sight to justify the conclusion drawn from it, but I think it may be otherwise accounted for. It is easy, of course, to understand that, if the nurses and servants were properly revaccinated before entering the institution, they would be at the time perfectly protected against small-pox. In proportion, however, as the immunity thus afforded by the vaccine virus is wearing itself out, so, *pari passu*, the small-pox virus from such a constantly infected atmosphere is supplying its place, but in such infinitesimal quantities that no characteristic symptoms are developed, and it thus becomes its own protective agent. If this explanation be correct, it is easy to see that the assertion of Mr. Simon, founded on this experience alone, cannot be received as proved.

I may mention here a theory bearing on the subject, which I dare say is possessed by many others besides myself, that the reason why men of our profession are proverbially insusceptible of fevers is, that they are in the habit of imbibing from time to time small quantities of any particular fever-poison, not sufficient to produce an attack of fever, but giving rise, perhaps, to temporary headache, sore throat, or other slight symptom; this gradually renders the system incapable of multiplying the fever-poison, the capacity for it being exhausted, just as one attack of one eruptive fever gives considerable immunity from another.

DISCUSSION IN THE SURGICAL SOCIETY OF IRELAND.

At a meeting of the Surgical Society of Ireland, on Friday, March 24th, Albert J. Walsh, Esq., President, in the Chair, Dr. CHARLES F. MOORE read a paper on Vaccination and Revaccination. The author, in performing the primary operation, preferred two sets of very slight incisions, or rather scratches made with a sharp lancet. Lymph might best be obtained from a healthy infant aged between three and six months. In cases where it was deemed inadvisable to puncture the lymph vesicles, removal of the thin crusts or scabs generally formed in the line of the incisions would usually afford abundance of lymph. If variola were epidemic, Dr. Moore was in favour of vaccinating infants of very tender age, even those but a fortnight old. Though primary vaccination was almost universally performed in Ireland, yet immigration from England was constantly introducing a number of unvaccinated persons into this country. Instances of the importation of small-pox in the same way by isolated individuals during the present epidemic were alluded to. Among these was the case of a girl, E. C., who, seventeen days after leaving London, was seen by Dr. Moore in incipient variola. She was at once sent to Hospital, where, in six days, she succumbed to the disease. This, it may be mentioned, has been the only instance in which small-pox has terminated fatally in Dublin since the outbreak of the present epidemic. The experiences of various Small-pox Hospitals in London, of the Public Services, and of numerous private medical authorities were quoted, as showing the protection to life afforded by vaccination, and the existence at present of an unusual susceptibility to vaccinia. Aware that some, whose opinion he respected, considered the primary operation sufficient, Dr. Moore was yet constrained to regard revaccination as most useful, especially in times of a variolous epidemic. In the cases he had met with, any tendency to excessive inflammation had been successfully combated by the adoption of ordinary precautions against rubbing or other irritation. The remarkable exemption from small-pox enjoyed by the Army, the Navy, and the Customs Department must be looked upon as in great measure due to the general practice in those services of revaccination. No doubt, as was demonstrated by drawings made by the author and exhibited to the meeting, in the second operation an earlier maturation of the vesicles occurred as compared with primary vaccination, yet this did not militate against the advantages of that second operation. In cases where only one set of incisions takes after primary vaccination, Dr. Moore was in the habit of performing auto-revaccination. He concluded by expressing himself as entirely in favour of the second operation.—Mr. TUFNELL brought forward a *résumé* of nearly 1,000 cases of revaccination, 90 of which had been performed at the Military Hospital at Kilmainham. The remainder were collected from various military authorities. In none of these 1,000 cases did any ill result follow the operation, and in many of them the vesicles were as perfectly formed as if they had been primary cases. In his experience, even the severity of prison drill had had no injurious effect upon the revaccinated.—Dr. CRONYN expressed himself as opposed to the second operation, on the ground that, in his experience, ill effects often were its

direct consequence. Further, he had not seen a perfect vaccine vesicle after it.—Dr. MCCLINTOCK had not seen injury done by revaccination. He thought the question as to the duration of the protective influence of primary vaccination one of great importance. The occurrence of a perfect revaccination vesicle surely warranted the conclusion that the individual was susceptible to an attack of variola. At the same time he believed that if the operation were once *well* performed it was unnecessary to repeat it.—Dr. DARBY did not approve of revaccination, inasmuch as he had seen severe ill-consequences result from it, and as the lymph derived from it was, in his opinion, useless.—Mr. MADDEN, Surgeon-Major, said that of 1200 cases of revaccination which he had performed among the troops under his care, none had had to be sent to hospital, or even to leave off drill for a single day.—Dr. ATTHILL considered that the efficacy of revaccination had been established beyond all dispute by the recent stamping out of an outbreak of variola in St. George's Hospital. He did not think that a good mark insured protection more than an ill-defined one.—Dr. ROBINSON had never seen variola in a revaccinated individual.—Mr. CROLY spoke in support of the second operation. In his opinion, the inflammation which occasionally followed revaccination was merely accidental, and, as such, comparable to the ill effects that in certain instances are apt to follow the infliction of ever so trivial a wound.—Mr. WHITE did not approve of a repetition of the operation, and he had never yet revaccinated an individual.—Dr. FREDERICK ROBINSON, Surgeon Scots Fusilier Guards, stated that the men joining his regiment were invariably revaccinated, and that, though in London they had been more exposed to the contagion of small-pox than most troops, the cases of the disease among them were extremely few. Idiosyncrasy may outweigh even revaccination. The officers had all lately been revaccinated, and scarcely one had suffered from the operation.—Dr. THOMAS BEATTY had had an average of 60 primary vaccinations in private practice annually for the last forty years. He had never heard of any of these having been attacked by variola subsequently. He approved of revaccination as a matter of expediency, but did not regard it as necessary. The age most suitable for its performance was 20 years.—Dr. CHURCHILL, from an experience extending over thirty-nine years, and based upon 2000 cases under his own observation, agreed with Dr. Beatty's views on the subject. He believed the question, whether liability to the poison of variola had necessarily existed before revaccination in a given case, to be still unanswered.—Dr. EVORY KENNEDY asserted that it was only the result of accumulated observations which could lead to a reliable opinion. Vaccination afforded but a partial protection against the variolous poison. This protective influence wore out in time, but could be renewed. He confirmed Dr. Atthill's observations respecting St. George's Hospital by adducing a similar group of circumstances, and he expressed himself altogether in favour of revaccination where there was exposure to the contagion of variola. The debate was then adjourned, and the meeting separated.

VACCINATION.

SIR,—A public vaccinator asks, what are the powers of public vaccinators as to taking lymph from healthy children? Your reply that there is no legal power to compel this, is undoubtedly correct, but your fear that parents may make this portion of the Act a dead letter, is, I think, groundless. With the experience of many years, vaccinating more than a thousand children yearly, I have never met with any serious difficulty or obstruction in that direction. I should regret any attempt to enforce submission by law, to a proceeding which in my judgment can be effected more agreeably, more satisfactorily, and, I believe, also with much greater facility by tact, kind persuasion, and a judicious explanation of the object, than by any legal enactment.

If Dr. Woodward desires confirmatory evidence of the manifest advantage of the concentration of public vaccination, he will find ample proof in the reports issued by the Medical Officer of the Privy Council; the history of the public vaccination of this country has placed the question beyond controversy. I have recently experienced the inconveniences of subdivision at my own station, where, during the small-pox panic and revaccination mania, it was necessary to vaccinate twice a week instead of once; so long as the numbers applying were double the usual average, all went on well, but when the pressure subsided and the numbers fell to the ordinary rate, I have been obliged to return to one day a week, in order to secure a proper selection of lymph.

I am, etc.,
March 25th, 1871.

A. B. STEELE,
Member of the National Vaccine Establishment.

USING AN OPPORTUNITY.

At a recent meeting of the Croydon Local Board of Health, when the report of the Sanitary Committee was brought up, Dr. Carpenter took the opportunity of addressing the Board—and through them the public—on small-pox and vaccination. He commented especially on an article in the *City Press*, in which the value of vaccination was disparaged, a statement being made that it appeared no longer to be the safeguard that it had been. In refutation of this, Dr. Carpenter adduced the instances of the medical staff and nurses of St. George's Hospital, and of the staff of the London Small-pox Hospital; as well

as the effects of small-pox among vaccinated and non-vaccinated persons in Croydon. Other popular objections against vaccination were also confuted. Dr. Carpenter has done good service in thus contributing towards the enlightenment of the public in a matter regarding which they are too liable to be misled by obstinate stupidity. We are glad to see his remarks printed at length in a local paper.

SPECIAL CORRESPONDENCE.

VIENNA.

[FROM OUR OWN CORRESPONDENT.]

Unveiling of the Portrait of Skoda.

THIS morning, at eleven o'clock, the portrait of Professor Skoda was unveiled in the lecture-room adjoining his clinical wards. The professors, students, many physicians, and the members of the Academy of Sciences (all in full dress) took part in the ceremony, which was very simple, but imposing. Professor Dlauhy—Skoda's oldest friend—delivered a short congratulatory speech, after which the portrait of the retiring professor was unveiled. The picture is a life-size one, and represents Skoda standing with his right hand in his pocket, as he usually keeps it while he is lecturing. As a likeness, it is extremely good, fully impressing one with the idea that one is looking on the real and peculiarly striking features of Skoda, and might expect to hear him begin his lecture over the patient. It is the work of Gustav Gaul, one of the most famous Austrian artists now living.

Immediately after this ceremony, deputations of numerous societies met at the house of Skoda, to present him with their various addresses. Here Professor Hebra delivered a highly characteristic speech, which was much applauded, introducing at the same time, side by side, the different deputations. These were: from the Academy of Sciences, speaker, Professor Arneht; from the Professor's College, speaker, Dr. Braun; from the College of Physicians of Vienna, speaker, Dr. Schlesinger; from the Society of Physicians, speaker, Professor Rokitansky; from the Medical Club, speaker, Professor Benedikt; from the medical staff of the General Hospital, speaker, Dr. Hofmann; from the Wiedner Hospital, speaker, Dr. Dinstel; from the Rudolf Hospital, speaker, Dr. Böhm; a deputation from the Royal Society of Physicians of Pesth, speaker, Prof. Lumniczky; and a deputation from Franzensbad, which presented the professor with the freedom of their city.

Skoda was so much affected that he could scarcely speak, and declared himself totally unable to reply to the various allusions made by Hebra, or to thank him for the numerous compliments paid to him. He was, he said, really astonished to find himself inundated by so much love and honour. He thanked all in simple words, saying that the credit of having raised the Vienna Medical School to its present position belonged only in a very small part to himself; that the larger part belonged to Rokitansky, to many other scientific men who were now amongst those around him, and to others now dead. He also detailed the reasons which induced him to retire. He said that he could now no longer participate in the progress of medical science, because his failing sight prevented him from reading; and that a man who could not follow the advances of science "was not worthy of being a professor."

Vienna, March 25th, 1871.

THE POOR-LAW MEDICAL SERVICE

OF GREAT BRITAIN.

THE CORONERS' BILL.

AT a meeting of the Poor-law Medical Officers' Association on Tuesday, a resolution was unanimously passed, condemning clause vii of the Coroners' Bill, introduced by Mr. Goldney, M.P., and Mr. Thomas Chambers, M.P. This clause was framed to deprive Poor-law Medical Officers of Workhouses of their fees for making *post mortem* examinations and attending coroners' inquests to give scientific evidence. As the Bill has since been withdrawn, we will only remark that the clause would have, if passed, inflicted a palpable injustice on the Poor-law medical service, while, at the same time, it carried with it the latent means of suppressing inquests in our Poor-law establishments.

VACANCIES.

ALNWICK UNION, Northumberland—Medical Officer for the Workhouse, and Medical Officers and Public Vaccinators for the Alnwick and Lesbury Districts.
AYSGARTH UNION, Yorkshire—Medical Officer and Public Vaccinator for the Askrigg District.

CREDITON UNION, Devonshire—Medical Officers for the Bow and Colebrook Districts.
GLENMUICH, Aberdeenshire—Parochial Medical Officer.
HENLEY UNION, Oxfordshire—Medical Officer for the Nettlebed District.
LANGPORT UNION, Somersetshire—Medical Officer for the Workhouse.
ORMSKIRK UNION, Lancashire—Medical Officer for District No. 4.
WOBURN UNION, Bedfordshire—Medical Officer for the Toddington District.

THE POOR-LAW MEDICAL SERVICE OF IRELAND.

WE are happy to find that our recent comments on the injustice of the law affecting Irish Poor-law medical officers in respect to the certification of dangerous lunatics is likely to be followed by action arising out of those comments in the sense there indicated. The action to be taken by the Irish Poor-law Medical Officers' Association will, we are happy to find, be aided by that of the English Association at the proper time. The Parliamentary Committee of the British Medical Association will bring its influence to bear in the same direction.

GRATUITOUS PSYCHOLOGY.

IN Ireland, the dispensary medical officer is called upon to certify to lunatics without "fee or reward". The performance of this duty is always attended with considerable trouble and loss of time; and not infrequently the medical officer has to pay money out of his own pocket for travelling expenses. The English Poor-law Medical Officers' Association have drawn up a petition, which will be presented to Parliament, praying that this grievance of their Irish brethren may be redressed. It is satisfactory to know that several eminent members of Parliament have expressed their intention of supporting the petition. It is not a little remarkable that, when medicine impinges on law, lawgivers should so frequently take the occasion for attempting to despoil the profession of medicine of its legitimate rewards.

VACANCIES.

CAHERCIVEEN UNION, co. Kerry—Medical Officer for the Emlagh Dispensary District.
CALLAN UNION, co. Kilkenny—Medical Officer, Public Vaccinator, and Registrar of Births, etc., for the Kilmoganny Dispensary District.
WESTPORT UNION, co. Mayo—Medical Officer for the Islandeady, Westport, and Louisburgh Dispensary Districts.

ASSOCIATION INTELLIGENCE.

BATH AND BRISTOL BRANCH.

THE fifth ordinary meeting of this Branch will be held at the Royal Hotel, College Green, Bristol, on Thursday, April 13th, at 7 P.M.; CHARLES BLECK, Esq., President, in the Chair.

Papers are promised by Dr. E. L. Fox, Dr. W. Budd, Mr. Leonard, Mr. Tibbits, and Mr. Dowson.

R. S. FOWLER, } *Honorary Secretaries.*
E. C. BOARD, }

Bristol, March 29th, 1871.

METROPOLITAN COUNTIES BRANCH.

AN ordinary meeting of this Branch will be held at the rooms of the Medical Society of London, 32A, George Street, Hanover Square, on Friday, April 21st, at 8 P.M.; when Dr. E. C. SEATON will open a discussion on Some of the Lessons to be derived from the present Epidemic of Small-pox.

A. P. STEWART, M.D. } *Honorary Secretaries.*
ALEXANDER HENRY, M.D. }

London, March 29th, 1871.

CUMBERLAND AND WESTMORLAND BRANCH.

THE spring meeting of the above Branch will be held at Kendal, on Wednesday, May 3rd, 1871. THOMAS F. I'ANSON, M.D., President in the Chair.

Gentlemen intending to be present, or to read papers, are requested to communicate with the Secretary without delay.

HENRY BARNES, M.D., *Honorary Secretary.*
Carlisle, March 29th, 1871.